

Kinematic Features (x_k)

$$\begin{aligned} q &\in \mathbb{R}^5 \quad (\text{Positions, rad}) \\ \dot{q} &\in \mathbb{R}^5 \quad (\text{Velocities, rad/s}) \\ \ddot{q} &\in \mathbb{R}^5 \quad (\text{Accelerations, rad/s}^2) \\ \tilde{x}_k &= [\tilde{q}^\top, \tilde{\dot{q}}^\top, \tilde{\ddot{q}}^\top]^\top \in \mathbb{R}^{15} \end{aligned}$$

Target (Ground Truth)

$$\begin{aligned} \tau^* &\in \mathbb{R}^5 \quad (\text{Measured joint torques, N-m}) \\ \tilde{\tau}^* &= (\tau^* - \mu_\tau) / \sigma_\tau \quad (\text{Normalized}) \end{aligned}$$

Total Analytical Torque τ_{phys}

Rigid-Body Dynamics (RNEA)

$$\begin{aligned} \tau_{rnea} &= \tau_M + \tau_C + \tau_g \in \mathbb{R}^5 \\ \tau_M &= M(q)\ddot{q} \quad (\text{Inertia}) \\ \tau_C &= C(q, \dot{q})\dot{q} \quad (\text{Coriolis}) \\ \tau_g &= g(q) \quad (\text{Gravity}) \end{aligned}$$

Friction Model

$$\begin{aligned} \tau_f &= c \odot \tanh(\dot{q} / \varepsilon_f) + v \odot \dot{q} \\ c &: \text{Coulomb coeff. (N-m)} \\ v &: \text{Viscous drag coeff.} \\ \varepsilon_f &: \text{Stiction half-width} \end{aligned}$$

+

$$\tau_{phys} = \tau_{rnea} + \tau_f = \tau_g + \tau_M + \tau_C + \tau_f$$

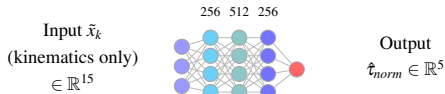
Shared Hyperparameters & Experimental Grid

Architecture: MLP [256, 512, 256], SiLU, LayerNorm
Dropout: 0.4 (Model II) / 0.1 (Models III & IV)
Optimizer: AdamW ($\eta_0=3 \times 10^{-4}$, $WD=5 \times 10^{-3}$), Warm-up Cosine
Max Epochs: 1000 (II) / 100 (III & IV), Early Stop patience 50–80

Experimental Grid (Models III & IV, 2 seeds each)

Data fractions (f): {1.0, 0.5, 0.25, 0.1, 0.05, 0.02}
Physics weights (α_0): {0.05, 0.1, 0.2, 0.5, 1.0, 2.0} (Model III)
Correction penalty (α_{reg}): {0.005, 0.01, 0.05, 0.1, 0.5} (Model IV)

(A) Black-Box FNN



Architecture

$$\hat{\tau}_{norm} = f_\theta(\tilde{x}_k) \quad (\text{plain MLP})$$

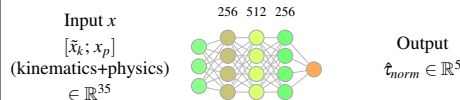
Input–Output dims: 15 $\rightarrow$ 5

Training Objective

$$\mathcal{L} = \mathcal{L}_{data} = \frac{1}{N} \sum_{i,j} w_j (\hat{\tau}_{norm,ij} - \tilde{\tau}_{ij}^*)^2$$

$$w = [1.0, 2.5, 1.0, 1.0, 1.0]^\top \quad (\text{J2 upweighted})$$

(B) Physics-Regularized FNN



Architecture

$$\hat{\tau}_{norm} = f_\theta(x) \quad (\text{augmented MLP})$$

$$\text{Input–Output dims: } 35 \rightarrow 5$$

$$x_p = [\tilde{\tau}_g^\top, \tilde{\tau}_M^\top, \tilde{\tau}_C^\top, \tilde{\tau}_f^\top]^\top$$

$$\tilde{\tau}_{k,j} = (\tau_{k,j} - \mu_{\tau,j} / 4) / \sigma_{\tau,j}$$

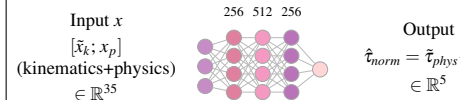
Training Objective

$$\mathcal{L} = \mathcal{L}_{data} + \alpha_{eff}(e) \mathcal{L}_{phys}$$

$$\mathcal{L}_{phys} = \frac{1}{N} \sum_{i,j} w_j (\hat{\tau}_{norm,ij} - \tilde{\tau}_{phys,ij})^2$$

$$\alpha_{eff}(e) = \alpha_0 \cdot \min(1, e / [0.05 E]), \quad \alpha_0 = 0.5$$

(C) Physics-Residual PINN



Architecture (PINN)

$$\Delta\tau = \lambda \cdot \tanh(\text{MLP}_\theta(x)), \quad \lambda = 0.5 \text{ (fixed)}$$

$$\hat{\tau}_{norm} = \tilde{\tau}_{phys} + \Delta\tau, \quad \|\Delta\tau\|_\infty < \lambda$$

$$\text{Init: } W^{(4)} \leftarrow 0.01 W_{\text{Xavier}}$$

$$\Rightarrow \hat{\tau}_{norm}|_{e=0} = \tilde{\tau}_{phys}$$

Training Objective

$$\mathcal{L} = \mathcal{L}_{data} + \alpha_{reg} \mathcal{L}_{reg}$$

$$\mathcal{L}_{reg} = \frac{1}{N \cdot J} \sum_{i=1}^N \|\Delta\tau_i\|^2, \quad \alpha_{reg} = 0.05$$

Penalizes large corrections; encourages minimal departure from τ_{phys}